

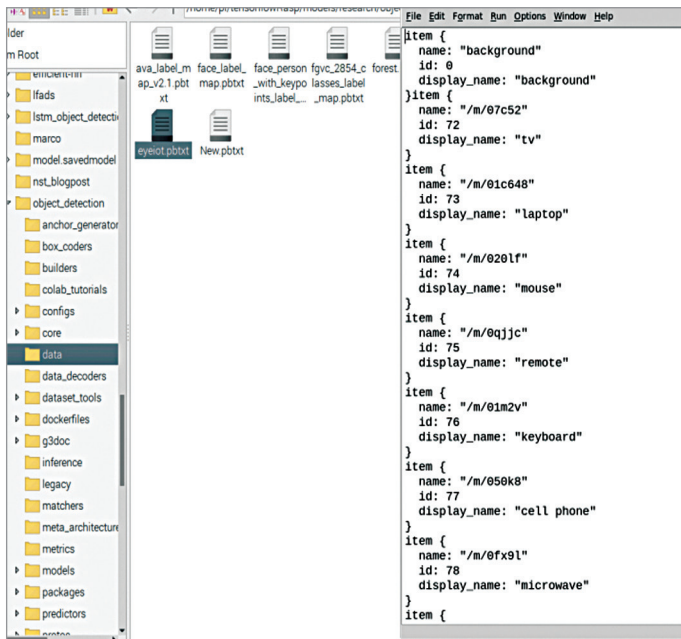


Fig. 4: The .pbtxt file having list of appliances and devices to detect

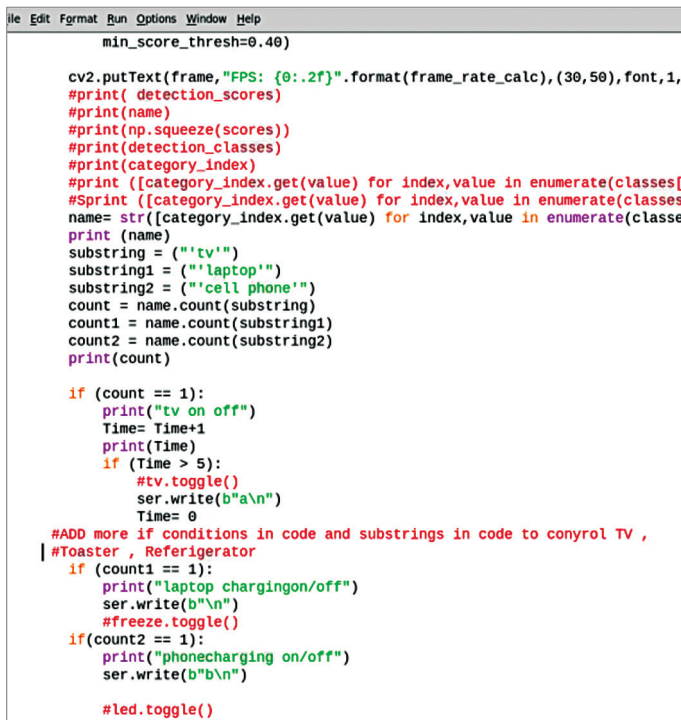


Fig. 5: Code detecting the device user is looking at

To create the IoT object detection and controlling functionalities, copy the code for object detection to a new file. Make sure to import a few more modules and libraries to the code such as gpiozero to control the pin inputs and outputs and pyserial to send the command to Pico to control the relay and thus the appliances.

We use pyserial module to connect to RPi Pico using Bluetooth HC 05 attached with RPi Pico and built-in Bluetooth of Raspberry Pi 4. So, in the code import the py serial and set the baud rate and serial name of Bluetooth serial port

Finally, set the file name `eyeiot.pbtxt` in the code so that it can detect

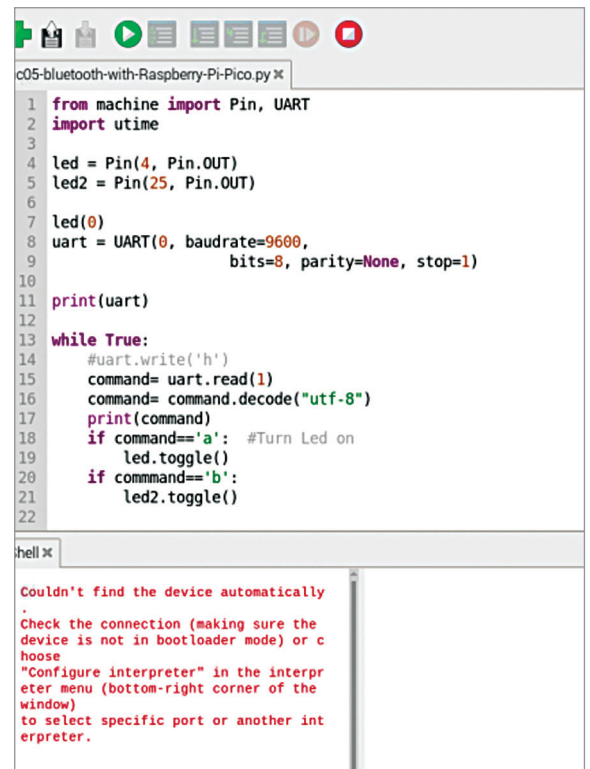


Fig. 6: Raspberry Pi Pico code to control the device

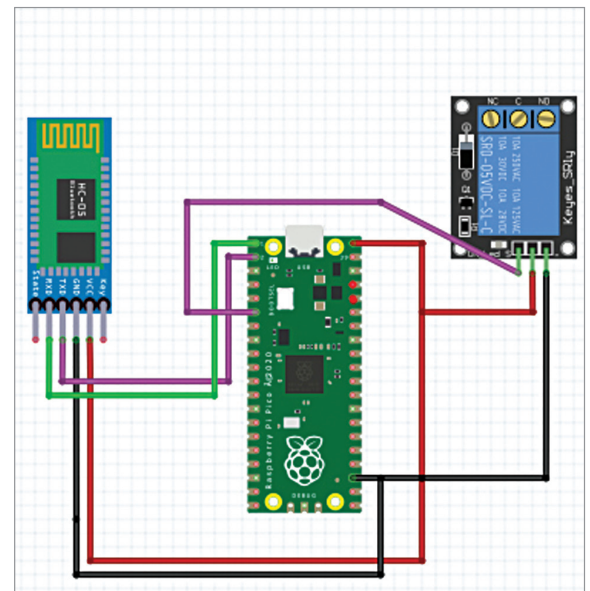


Fig. 7: Connection of Rpi Pico with Bluetooth and relay module

which appliance you wish to control from the given list of appliances (whether it is refrigerator, TV, toaster, or any other). After doing so, obtain the result where the strings contain the list of detected object names on the video.

The code will search various sub-string names in the string format